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A reference architecture for the operationalization of machine learning models in manufacturing

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Abstract

Inherent characteristics of machine learning algorithms such as their probabilistic nature, their reliance on large datasets as well as their need for constant retraining pose major challenges to the operationalization of machine learning models (MLOps) in the manufacturing domain. As such systems are known to quickly accumulate technical debt due to system-level interdependencies of code, data, and models, clear abstractions boundaries are mandatory. Therefore, this publication derives a systematic functional decomposition of an MLOps system tailored to the manufacturing industry into specific domains and contexts. Moreover, a concrete deployment view is provided, and possible future research directions are discussed.

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1. Motivation

Smart Manufacturing is among the most influential trends in modern manufacturing. The amalgamation of information technology (IT) and operational technology (OT) in a so-called cyber-physical system (CPS) enables new opportunities regarding flexibility, speed, and robustness in manufacturing automation [1,2]. Promising approaches to autonomous CPS are data-driven techniques such as Machine Learning (ML) or Deep Learning (DL) potentially allowing the efficient processing of data acquired by the CPS's sensors [3].

However, while recent efforts of Machine Learning in Industrial Internet of Things (IIoT) scenarios mainly focus on the individual CPS, little research is conducted on the integration of distributed CPS across manufacturing lines to establish cyber-physical production systems (CPPS) [3]. The demand for vertical and horizontal integration of individual CPS as well as other upstream assets is further amplified when considering the requirements and potential use cases of ML models in a manufacturing environment. ML algorithms can model process interdependent use cases such as predictive quality as well as the adaption of process parameters, and thus demand integrated data acquisition and analysis across the process chain. Moreover, in contrast to the training phase when huge computational efforts are required, ML models are less resource-intensive in production for which low latencies are paramount to adhere to strict takt times [3].

This emphasizes the future importance of Continuous Delivery (CD) of centrally trained ML models and supporting services in a decentralized manufacturing environment as well as their use-case-driven integration and combination in the CPPS. Hence, an architecture that combines cloud and edge resources into one system is required. Hereby, ML makes high demands with regards to the integration of industrial PC (IPC)

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close to the manufacturing process, the required data, the edge service integration, DevOps approaches such as CD, Continuous Integration (CI), Continuous Training (CT), and ML models development cycle. Therefore, a reference architecture for Machine Learning Operations (MLOps) in manufacturing is deducted within this publication.

Consequentially, the paper is structured as follows. In chapter 2 the necessary fundamentals of ML and MLOps are presented. In chapter 3, challenges for ML systems in manufacturing are discussed. Chapter 4 conducts a functional decomposition of an MLOps system in the domain of manufacturing. The paper concludes by discussing and highlighting future research opportunities.

2. Fundamentals and state of the art

2.1. Machine Learning

ML algorithms derive decision patterns from preprocessed data (feature engineering) to solve a given problem without being explicitly programmed [4]. Moreover, as a subcategory of ML, DL further automates data analysis by combining feature engineering and decision-making in an architecture that is trained end-to-end and excels at the analysis of unstructured, high-dimensional data such as time series and images [5]. Depending on the task and available data, most ML applications can be further categorized into supervised, unsupervised, and reinforcement learning [6].

Developing ML models is a time-consuming process and inherently uncertain whether the initial stated goals can be achieved. According to the standard process model CRISP-DM (Fig. 2), the process is highly iterative [7].

Fig. 1. CRISP-DM steps. [7]

2.2. Machine Learning Operations

MLOps comprises technological capabilities and tools for developing and operating ML systems efficiently at scale. As MLOps emerged from DevOps paradigms, Continuous Delivery (CD) best practices and approaches are applicable. These approaches include version control, deployment and test data management, test automation, monitoring, proactive notification, integrated security, and loosely coupled architecture [8].

In contrast to traditional software engineering, machine learning engineering differs primarily in the extensive integration of data, code, and models [9,10] as well as the experimental, often uncertain development process [11,12]. Concerning the ML model, additional effort is incurred for testing and monitoring [13,14], model management [15], and continuous deployment of the resulting model. Furthermore, the additional effort for data and model management also requires efficient configuration management [12,16].

An exemplary generic MLOps pipeline is given in Fig. 1 [17]. Central to the MLOps pipeline is data and model management that serves as a single contact point for different metrics acquired during development and training required for compliance, traceability, and auditability. The development of ML models relies heavily on high-quality data, thus dataset and feature repositories are mandatory to reduce the overall effort spend in feature engineering and reduce discrepancies between training and serving data. Developed ML pipelines and supporting code is then pushed to version control from which the automated training pipeline is derived, continuously updating trained models in a model registry. Subsequently, the updated model is then deployed into production using an optimized serving engine and continuously monitored for deteriorating performance or changes in the environment. Depending on predefined thresholds, retraining is triggered.

3. Challenges of ML systems in manufacturing

While CPS and ML are utilized across various industries,

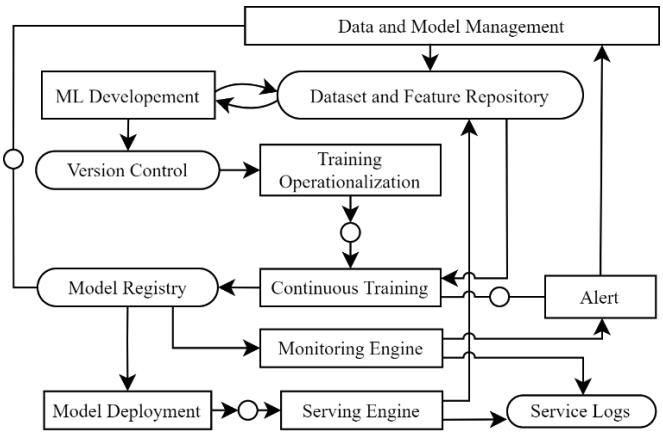

Fig. 2. Generic end-to-end MLOps pipeline. [17]

applications from adjacent industries such as energy, healthcare or transportation are not directly transferable to Smart Manufacturing [18], due to numerous influencing factors discussed below. Thus, to ensure maximum efficiency of the resulting MLOps architecture, the specific considerations of manufacturing need to be taken into account. This publication subdivided the challenges identified in the literature into four subcategories that are discussed in detail in the following:

- **Heterogeneity:** Manufacturing systems are highly heterogeneous environments. Hereby, challenging aspects include the spatio-temporal distribution of operational technology (OT) and informational technology (IT) assets [19], inhomogeneous communication protocols [20], machinery from various vendors, diverse manufacturing processes [18], a variety of expected ML use cases and modeling techniques [21–23], different data modalities [24] and potentially high numbers of product varieties [25].
- **Data Management**: Data in adequate quality and quantity is fundamental to the performance of ML models [26]. In manufacturing, data is acquired in structured, semi-structured, and unstructured formats [24]. This multi-modal data results in significant

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challenges for DL architectures [26] such as Convolutional Neural Networks (CNN) or long short-term memory (LSTM) that introduce data-specific

- inductive biases into the architectures [5]. Moreover, as the performance of supervised learning approaches deteriorates when faced with imbalanced data [27], the management of faulty data is especially important [25,26]. Finally, the flexible integration of various data sources such as process data, machinery data, and management data is also a major concern [28].
- **Operations**: Since the validity of an ML model is tied to the environment it was trained in, changes to it might affect its performance [29]. Manufacturing systems are subject to sudden changes due to small batch manufacturing, human intervention, or process drift [30] which pose significant challenges for ML. This necessitates constant monitoring and incremental learning. Additionally, to foster trust in the ML model's decision by the workers, interpretability is crucial [25]. Moreover, potentially high cycle times require very short latencies as compliance to short takt times is mandatory [31].
- **Non-Technical**: As an innovative and interdisciplinary technology, ML systems suffer from uncertain terminology and expectations of manufacturers, thus limiting the successful application [29]. Moreover, cost sensitivity is also a major concern to justify the application of ML [25]. Lastly, skills shortage and qualification issues [32] of manufacturing personnel also need to be addressed to allow for scalable solutions.

4. A reference architecture for MLOps in manufacturing

In Software Engineering (SE), modular and encapsulated architectures enable the development of maintainable and flexible software. ML systems especially are difficult to decompose, as ML models tend to erode abstraction boundaries due to their dependencies on system-level behavior such as changes in acquired data [33]. Thus, to develop a holistic MLOps reference architecture for manufacturing, this work adopts the Strategic Design approach from domain-driven design (DDD) [34]. DDD aims at identifying expedient abstractions and encapsulations, and thus, is especially suitable for the development of software in interdisciplinary and complex domains.

4.1. Domain View

In this chapter, a domain model (Fig. 3), representing the problem space, is derived to address the prior identified requirements. A domain model representing the business context consists of *subdomains* and *bounded contexts* [34].

The domain model encompasses five bounded contexts – *Edge system, cloud instance, data management, development,* and *MLOps*. From an architectural perspective, this represents the conceptualized cloud edge architecture to account for the decentralized characteristic of manufacturing shopfloors as well as strict takt time requirements. Moreover, clear abstraction of subdomains provides guidelines for an expedient

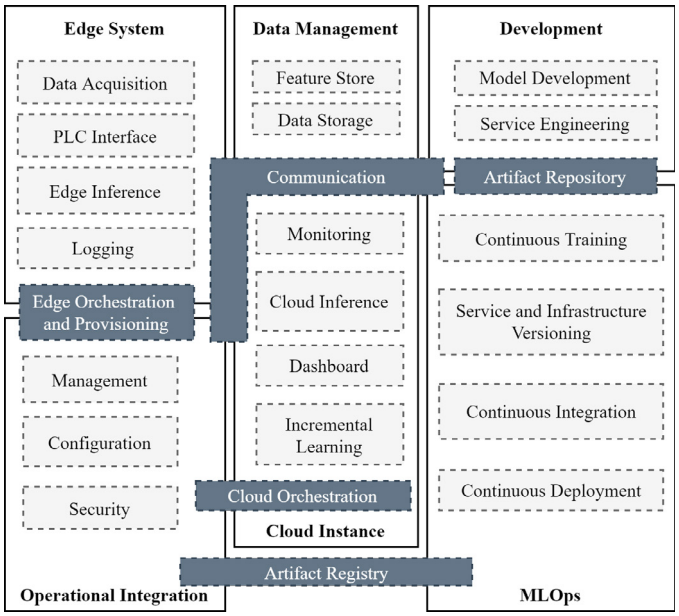

Fig. 3. Domain model of an MLOps system for manufacturing showing bounded contexts (solid lines) and subdomains (dashed lines).

containerized microservice architecture. This can be beneficial to achieving the required flexibility towards communication protocols, use cases, manufacturing processes, etc. Furthermore, the communication subdomain to connect dynamically changing and distributed OT and IT assets is a key area to focus on.

The *edge system* is required to perform time-sensitive and data quality-related tasks such as interacting with data sources and programmable logic controller (PLC) as well as providing local inference of trained ML models. Activities on the edge need to be closely monitored, thus, logging capabilities are extremely important.

In contrast to the time-critical applications at the edge, *cloud instances,* hosted on central computing infrastructure, provide functionalities that feature a more distant time horizon. Among others, those functions are Monitoring, Dashboarding, Incremental Learning, and time-insensitive ML inferences (e.g. predictive maintenance).

The requirements regarding *data management* are twofold. Firstly, cost-effective long-term data storage for semi-structured and unstructured data is mandatory to provide high-dimensional data in sufficient quantity for model training. Furthermore, interfaces or adapters to structured databases, often for higher-level IT systems such as MES or SCADA, are necessary. Secondly, so-called cached feature stores can provide near-real-time access to aggregated features from structured databases or indexed semi-/unstructured data along the manufacturing line for individual workpieces.

According to SCULLEY et al. [33], ML applications require not only an ML model but require support from traditionally-coded software and configurations. Therefore, the *development* process of ML applications needs to incorporate appropriate development process models for ML and SE. Interfacing between the context *development* and *MLOps,* the subdomain *Artifact Repository* provides the developed architecture and further code artifacts for the subsequent CT/CI/CD.

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The *MLOps* context represents the architecture's capability to continuously deploy models that are trained on the latest data from the ongoing manufacturing process. Therefore, continuous as well as on-demand training of model artifacts combined with the ability to version the models, services, as well as configurations in a repository is necessary. Subsequently, upon changes, CI containerizes the software artifacts, performs extensive unit and functional tests, and commits the container to an artifact registry. Afterward, the CD is responsible to deploy the containers either to an edge or cloud device.

IT governance is fundamental for organizations to ensure the efficient and effective utilization of IT systems. As a subcomponent of IT governance, operational integration manages the interdependencies of organizational as well as IT infrastructure and processes [35]. Thus, the architecture's *operational integration* context is a central capability that manages and orchestrates the system's components as well as the communication. The management of spatio-temporal distributed edge devices, cloud instances, training pipelines, and configurations requires sophisticated orchestration capabilities. Moreover, security, access control, and data privacy are mandatory.

4.2. Deployment view

Based on the presented domain view (Fig, 3), given in Fig. 4, this chapter proposes a deployment view as a possible solution space. Based on the generic MLOps pipeline presented in chapter 2.2, the architecture is developed to satisfy the requirements of the domain model derived in chapter 4.1.

The edge device is the fundamental interface between OT and IT and employs an event-driven microservice architecture. Standardized interface services for sensors and PLCs are mandatory for scalable deployments. In terms of data acquisition, the data's quality is to be ensured during acquisition. Therefore, the Data Acquisition Services (DAQS) are tailored to the respective sensors and conduct data parsing, preprocessing, and transformation of the data into Industry 4.0 conform events [36] to be published to a lightweight central message broker (e.g. MQTT). Subsequently, the event is distributed to a local database for data persistence, a local cache for near-real-time access, and inference services that provide standardized and optimized interfaces to trained ML models. Besides the prediction, the inference service features options for logging and interpretability. Based on these predictions, decisions need to be derived and then acted upon by the Actor Interface. The Edge Connector is the communicative interface and gateway between the local and global event buses.

While the message broker at the edge enables inter-process communication and collaboration, the high-throughput global event streaming bus enables communication between different processes and cloud services. Moreover, this allows cross-process data analysis and ML use cases such as predictive quality. The acquired data at the edge is directly forwarded to the long-term data storage system and persisted according to the data's modality. As discussed in the previous chapter, a central feature store is used to calculate aggregated features from control IT systems and to allow near-real-time access to cross-process data for individual inference services.

Centrally hosted cloud services are time-insensitive services and applications that do not necessarily demand near-real-time latency. Most importantly, monitoring services to detect process drifts of incoming data, as well as concept drifts of deployed ML models, need to trigger retraining and deployment of models. Alternatively, Incremental or Active Learning is employed to continuously improve the performance of ML models and query workers if models are uncertain [37].

The development process of ML models utilized the traditional CRISP-DM. In parallel, supporting services such as DAQS, Actor Interfaces, and Decision Services need to be

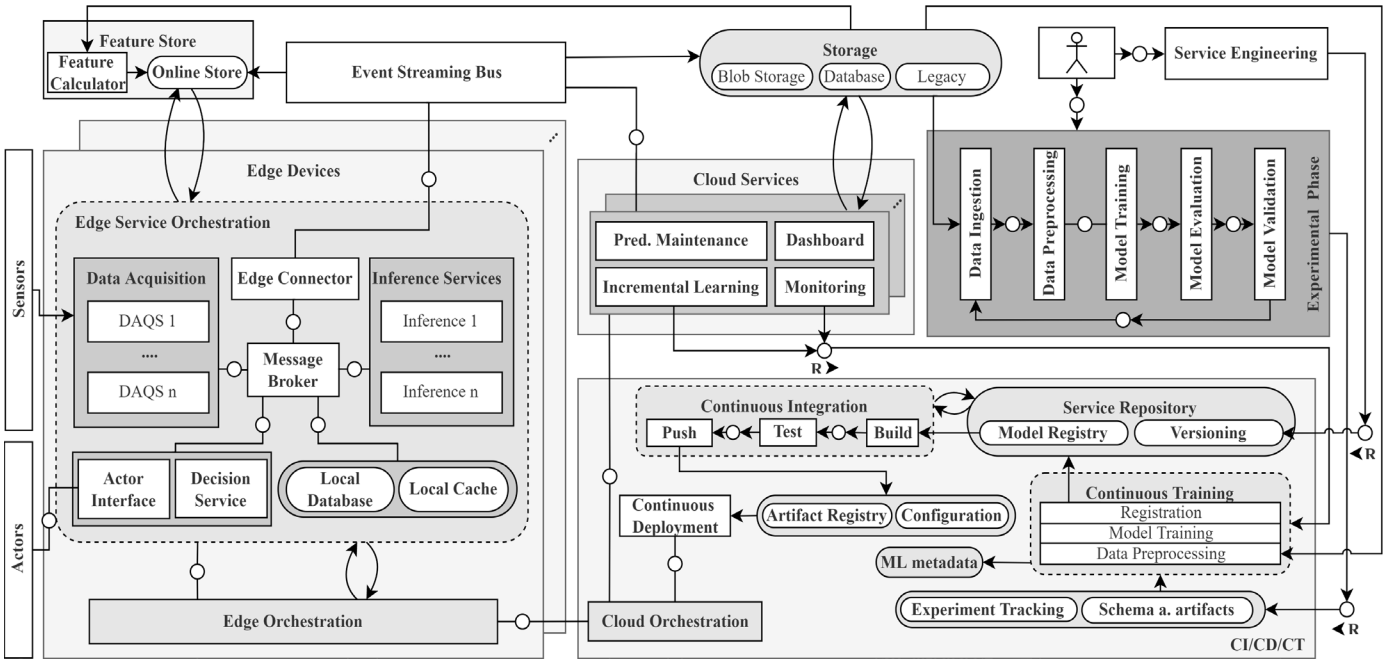

Fig. 4. Deployment view of the proposed reference architecture for MLOps in manufacturing (using the Fundamental Modeling Concepts Notation).

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developed. The development toolbox needs to be abstracted, standardized, and (at least partially) automated (e.g. Neural Architecture Search [31]) to allow scalable development in consideration of the skills shortage. Furthermore, data and resource-efficient approaches, as well as multimodal learning are to be deployed to minimize the amount of annotated data required as well as limit the number of architectures to be considered during modeling [38–40]. During the development of the ML models, extensive experiment tracking is mandatory. Finally, the best performing architecture is then versioned, whereby the models themselves, the configurations as well the data schemas are committed.

The CI/CD/CT pipelines represent the actual MLOps components in the architecture and are hosted on central GPU-enabled servers or in the cloud. The training pipeline (CT), constantly training the best performing ML models and saving training metadata for auditing purposes, is either triggered externally or interval-based. To achieve major cost savings for training, investments into efficient training pipelines are advised [41]. The trained models are then committed to the model registry in the Service Repository and brought together with its related services. Subsequently, upon a detected commit, the trained models or developed services are containerized, tested, and pushed to a container registry. Then, the CD pipeline deploys the latest containers to their respective edge device or cloud instance. Hereby, GitOps or Infrastructure as Code (IaC) paradigms [42] are especially relevant to achieving a stable, scalable, and reliable CI and CD process. Thus, it is recommended to utilize declarative configurable, hardware-agnostic container orchestration software such as Kubernetes as an orchestration engine [43]. In summary, Table 1 contrasts the possible technological solutions to the posed challenges in chapter 3.

Table 1. Synoptical comparison of possible solutions and challenges.

Challenges	Possible solutions
Heterogeneity	• Loosely coupling • Cloud edge architecture • Event-driven architecture • Standardized PLC and sensor interfaces • Containerization and orchestration • Neural architecture search (NAS)
Operations	• Holistic data acquisition • Optimized inference latency • Explainable AI • Drift and process detection • Metadata tracking • Active / Incremental learning
Data Management	• Standardized event messaging • Feature stores and caching • Multimodal data storage • Data-efficient training • Multimodal learning • Federated learning
Non-Technical	• Adequate, abstracted tooling • Optimized training pipelines • Infrastructure as Code (IaC) • Declarative configuration

5. Conclusion and outlook

This work derived a reference architecture for MLOps in manufacturing based on identified manufacturing-specific requirements as well as generic MLOps capabilities. Therefore, the business context is represented using a domain view and a possible solution is provided employing a deployment view.

Using MLOps, cost-efficient and scalable ML solutions enable intricate, cross-process data analysis to gain process knowledge about the effects of manufacturing processes on the materials as well as the resulting product.

Undoubtedly, due to numerous interdependencies and feedback loops, ML systems result in complex architectures. In this context, DDD is especially suitable to identify so-called Core Domains that are vital for the success of the intended business success. Consequentially, this domain is prioritized in terms of development effort.

As many business functionalities and services discussed in the previous chapters are already implemented and a plethora of open-source and commercial options exist [50, 51], this publication argues that the Core Domain is the architecture's operational integration layer. Due to cost sensitivity and skills shortage being major issues, increasing the development and deployment efficiency to enable scalable solutions can be a way to offset the initial investments and enable cost-effective applications of ML. However, as of today, data scientists are required to navigate a variety of data sources, model architectures, and tools while satisfying a multitude of use cases using a time-consuming development process. Therefore, developing a layer for the operational integration based on the models presented in chapter 3 to integrate the various edge devices, tools, databases, and dynamic components such as artifacts, models, services, configurations, or pipelines as well as managing their communication is necessary.

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